AN INTEGRAL DOMAIN EXTENSION OF A HENSELIAN LOCAL RING IS HENSELIAN LOCAL, AND STRICT HENSELIANNESS FORCES A PURELY INSEPARABLE RESIDUE EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span contains the author's three enumerated hypotheses, delivered verbatim. The author writes 'finite A-sub algebras' with a space inside the word; that is the author's text and is reproduced unchanged. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- A finite algebra over a henselian local ring is a product of local henselian rings. invoked at proof lines 30775-30776. Is the step that reduces the finite case to the domain case and gives locality.
- Going up for integral ring maps. invoked at proof lines 30778-30779. Locates the maximal ideal of B over that of A.
- Henselization and strict henselization of a local ring and the effect on the residue field. used at proof lines 30780-30784. Gives the purely inseparable residue extension and the strict henselian conclusion.

Source reference. Original lemma 04GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5834–5834. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04GH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5833–5833. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 620–620. Cross-chapter navigation pointer into the

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Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $A \to B$ be a ring homomorphism. Assume*

(1) *A is a henselian local ring,*
(2) *$A \to B$ is integral,*
(3) *B is a domain.*

Then B is a henselian local ring and $A \to B$ is a local homomorphism. If A is strictly henselian, then B is a strictly henselian local ring and the extension $\kappa(\mathfrak{m}_B)/\kappa(\mathfrak{m}_A)$ of residue fields is purely inseparable.

Proof. Write B as a filtered colimit $B = \operatorname{colim} B_i$ of finite A-sub algebras. If we prove the results for each B_i, then the result follows for B. See Algebra, Lemma 04GI. If $A \to B$ is finite, then B is a product of local henselian rings by Algebra, Lemma 04GH. Since B is a domain we see that B is a local ring. The maximal ideal of B lies over the maximal ideal of A by going up for $A \to B$ (Algebra, Lemma 00GU). If A is strictly henselian, then the field extension $\kappa(\mathfrak{m}_B)/\kappa(\mathfrak{m}_A)$ being algebraic, has to be purely inseparable. Of course, then $\kappa(\mathfrak{m}_B)$ is separably algebraically closed and B is strictly henselian. □